

Graduate Analysis 1: Assignment 1

Aditya Padiyar

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Definition 1. An outer measure μ on X is a monotone, countably subadditive map $\mu : 2^X \rightarrow [0, \infty]$ such that $\mu(\emptyset) = 0$

Definition 2. A countable union of nowhere dense sets is said to be a meagre set.

Definition 3. For $A \subseteq \mathbb{R}^n$, let $\mu_{\mathbb{Q}}(A)$ denote the number of rational points in A

Claim 1. Suppose $(X, \mathcal{F})$ is such that $X \in \mathcal{F}$ and $A \cap B^c \in \mathcal{F}$ for all $A, B \in \mathcal{F}$. Then $\mathcal{F}$ is an algebra on X

Proof. We must prove two things:

- Substituting X for A we get that $\mathcal{F}$ is closed under complements
- Now substituting B^c for B we get closure under binary intersection

□

Claim 2. Let $S \subseteq (0, 1)$ be defined by

$$S = \{x : \text{there is a decimal expansion of } x \text{ that contains the digit zero}\}$$

Then S is Borel and $\lambda(S) = 1$

Proof. Let

$$A_n = \{x : \text{some decimal expansion of } x \text{ has zero among the first } n \text{ digits}\}$$

Clearly we have $\lambda[(A_n)^c] = (\frac{9}{10})^n$. Now $\lambda(A_n)$ and since A_n increases to S we get $\lambda(S) = 1$

□

Claim 3. If μ_1 and μ_2 are outer measures on X then so is $\nu = \max(\mu_1, \mu_2)$

Proof. Let $A_n \subseteq X$ and $B = \bigcup_{i=1}^{\infty} A_n$. Then

$$\nu(B) = \mu_i(B) \leq \sum_{n=1}^{\infty} \mu_i(A_n) \leq \sum_{n=1}^{\infty} \nu(A_n)$$

□

Claim 4. *There exists $G, F \in \mathcal{B}[0, 1]$ such that*

- $G \cup F = [0, 1]$
- F is meagre in $\mathbb{R}$
- $\lambda(G) = 0$

Proof. Suppose $\{q_1, q_2, \dots\} = \mathbb{Q}$. Define

$$U_k = \bigcup_{j=1}^{\infty} (q_j - \frac{1}{2^{j+k+1}}, q_j + \frac{1}{2^{j+k+1}})$$

We have $\lambda(U_k) \leq \frac{1}{2^k}$ by countable subadditivity. Further, U_k is dense in $\mathbb{R}$ and hence $(U_k)^c$ is nowhere dense in $\mathbb{R}$. Define

$$G = [0, 1] \cap \bigcap_{k=1}^{\infty} U_k$$

then $\lambda(G) \leq \lambda(U_k)$ by monotone property of measure. Taking the limit, we see that $\lambda(G) = 0$. Now define

$$F = [0, 1] \setminus G = [0, 1] \cap \bigcup_{k=1}^{\infty} (U_k)^c$$

which is meagre in $\mathbb{R}$ □

Claim 5. *Suppose $f_n : [0, 1] \rightarrow \mathbb{R}$ are measurable. Then there exist a sequence of positive integers k_n such that $\frac{f_n(x)}{k_n} \rightarrow 0$ for almost all $x \in [0, 1]$*

Proof. Define

$$E_{m,n} = \{x : |f_n(x)| \geq \frac{m}{n}\}$$

For every fixed n , we have that $E_{m,n} \downarrow \emptyset$ and hence $\lambda(E_{m,n}) \rightarrow 0$ since the measure space is finite. Hence $\lambda(E_{k_n,n}) \leq \frac{1}{2^n}$ for some large k_n .

Define $A_n = E_{k_n,n}$ and $B = \limsup A_n$. We see that $\sum_{n=1}^{\infty} \lambda(A_n) \leq 1$ and hence $\lambda(B) = 0$ by Borel-Cantelli.

Now let $x \in B^c$. Then for $n \geq M$ we have $x \notin A_n$ which means $\frac{|f_n(x)|}{k_n} < \frac{1}{n}$ □

Claim 6. *Any Radon measure on $\mathbb{R}^n$ must be σ finite*

Proof. Define $K_m = [-m, m]^n$. We have $\mu(K_m) < \infty$ since K_m is compact. Since $\mathbb{R}^n = \bigcup_{m=1}^{\infty} K_m$ the measure μ is σ finite. □

Claim 7. $\mu_{\mathbb{Q}}$ is a σ finite measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$

Proof. We have that $\mathbb{R}^n = (\mathbb{R}^n \setminus \mathbb{Q}^n) \cup_{q \in \mathbb{Q}^n} \{q\}$. Each set in this union has finite measure. □

Claim 8. $\mu_{\mathbb{Q}}$ is not Radon

Proof. We have $\mu_{\mathbb{Q}}[0, 1]^n = \infty$, but a Radon measure must take finite values on every compact set $\square$

Claim 9. Suppose $\phi : X \rightarrow [0, \infty]$ is a measurable function on a σ -finite measure space X . Define

$$S = \{(x, t) \in X \times (0, \infty) : \phi(x) > t\}$$

Define $f : (0, \infty) \rightarrow [0, \infty]$ by $f(t) = \mu([S]^t)$ and $\mathcal{E} = \int_X \phi$. Then the following hold:

- f is measurable and $\mathcal{E} = \int_{(0, \infty)} f$
- $f(t) \leq \frac{\mathcal{E}}{t}$ (Markov's inequality)

Proof. We have the following simple facts

- f is monotone decreasing and hence measurable
- The map $\phi \times \text{id} : X \times (0, \infty) \rightarrow [0, \infty] \times (0, \infty)$ is measurable
- The set $\{(t_1, t_2) : t_1 > t_2\}$ is measurable in $[0, \infty] \times (0, \infty)$
- For all x we have $\int_{(0, \infty)} 1_{[S]_x} = \phi(x)$

Hence the set S is measurable and by Tonelli, we have

$$\mathcal{E} = \int_X \int_{(0, \infty)} 1_{[S]_x} = \int_{(0, \infty)} \int_X 1_{[S]^t} = \int_{(0, \infty)} f$$

Now fix $t \in (0, \infty)$. We have $t 1_{[S]^t} \leq \phi$. Integrating we get the desired result. $\square$